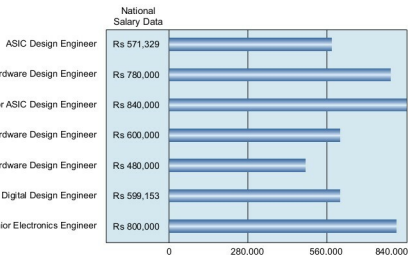


Most Popular Job for FPGA Skill



Note: Survey results according to Payscale.com

Infosys too have divisions for FPGA engineers.

This line offers creative and challenging work, which involves rapid prototyping of design ideas as well as verification of existing designs. Almost any area with design and verification tasks will have a requirement for FPGA-trained VLSI engineers to prototype the solutions of FPGA-based systems.

"FPGAs can be bought and programmed for a particular application. These are mostly used in embedded computing systems. All levels of companies—be it a manufacturer of high-end products like automobiles and cellphones, or low-end household equipment like coffee vending machines and toys—have FPGAs in their system," shares professor M. Balakrishnan, deputy director, department of computer science and engineering.

Balakrishnan explains with an example: "If a car manufacturing company wants to manufacture a new technology for a car, it will opt for an FPGA-based design for prototyping as it is available for \$5-10. It must have a good understanding of Verilog or VHDL language to design the slate, as FPGAs are a lot like a blank slate. The FPGA design engineer's responsibility lies in synthesising different

components inside and integrating them together."

A professor from IIT Bombay explains the role of an FPGA engineer: "A fresher's job will involve activities like testing, emulation and verification, while an expert in FPGA will be asked to handle digital system design, system-on-chip, embedded hardware-software co-design, digital communication sub-systems, hardware accelerators for various computing tasks, embedded control systems and digital signal processing sub-systems/systems."

Rajesh Choudhary, head of HR, Xilinx India, adds, "Their work spans across areas like software tool development, system design, IP design and IC design. They are expected to work in domains like bio-medical, video surveillance, embedded systems and communication system. They can grow into senior design engineers and project leads. Soft skills start coming into

The work profile of FPGA engineers spans across areas like software tool development, system design, IP design and IC design

the picture with managerial responsibility."

Must-have skills

Good digital electronics design skills, knowledge of programming platforms, a strong curriculum in VLSI design, good domain knowledge of digital signal processing or digital communications, computing architectures, control systems or appropriate engineering knowledge can take you a long way in getting hired by a good employer. Design work requires good understanding of the problem, not just programming or circuit

design skills.

Where to study

To get an entry to work at Xilinx, a bachelor's degree in electronics, electrical or computer science is compulsory. However, there is no pressure to take a postgraduation degree or specialisation for the same as the required topics are covered during graduation.

"We look for graduates from reputed institutes and we consider the university name very important while recruiting employees. We approach premium institutes like IITs and IIITs for placements as we have found from our experience that the quality of their talent is different. The two main factors we look for while recruiting people are technical capability in the subject and analytical power," shares Choudhary.

National Instruments works with some specific colleges for hiring FPGA engineers.

"Some of the institutes which we work very closely with are BITS Pilani and IIT Bombay, both of which have excellent VLSI design programmes. SJB Institute of Technology in Bengaluru is also an upcoming institute in FPGA-based graphical system design development," informs Karun Jain, senior technical consultant, National Instruments India.